

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=milk

retailer=silpo

promo_variant=promo_controlled

frequency=weekly

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

	series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p
producer_to_prozorro	producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	milk	silpo	promo_controlled	nan
	prozorro	nan	nan	nan	nan	nan	nan	ambiguous	0.0	milk	silpo	promo_controlled	nan
	retail	nan	nan	nan	nan	nan	nan	ambiguous	0.0	milk	silpo	promo_controlled	nan
	producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.9859002580259643
	prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.9859002580259643
	producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.9859002580259643

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
milk	silpo	promo_controlled	weekly	none	none	n/a	n/a	8	nan	nan	nan	nan	nan	nan	nan	unavailable	1	No admissible model estim...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
none	none	n/a	n/a	nan	nan	nan	1

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2025-10-27	9.391788400537894	44.1	59.99	milk	silpo	promo_controlled	daily_median_of_available...
2025-11-03	9.437110443078286	42.150000000000006	59.99	milk	silpo	promo_controlled	daily_median_of_available...
2025-11-10	9.541324852194801	42.0	59.99	milk	silpo	promo_controlled	daily_median_of_available...
2025-11-17	9.464479382910632	42.0	59.99	milk	silpo	promo_controlled	daily_median_of_available...
2025-11-24	9.351463254879393	42.84	59.99	milk	silpo	promo_controlled	daily_median_of_available...
2025-12-01	8.361658686536295	44.0	61.945	milk	silpo	promo_controlled	daily_median_of_available...
2025-12-08	7.848155395055493	43.8	59.99	milk	silpo	promo_controlled	daily_median_of_available...
2025-12-15	7.805102492916548	43.9	61.49	milk	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

No rows

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	any_i2	eligibility_note
milk	silpo	promo_controlled	weekly	ambiguous	ambiguous	ambiguous	0	Insufficient common sampl...

Table: NARDL_Multipliers

No rows

Table: VECM_IRF

No rows

time_series_promo_controlled.png

milk | silpo | promo_controlled

